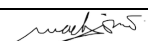


COURSE PLANNING (RPKPS)

UNIVERSITAS MULTIMEDIA NUSANTARA

VALIDATION PAGE

Course Name : Artificial Intelligence
Course Code : IF440
Course Coordinator : Moeljono Widjaja, Ph.D.
Team of Lecturers :

NO	NAME	NIK/NID	SIGN
1	MOELJONO WIDJAJA, PH.D.	068503	
2	YAYA SURYANA, PH.D.	L01172	
3			
4			

On behalf of the team,
Date:



Moeljono Widjaja, Ph.D.
Course Coordinator

Approved by
Date:



Arya Wicaksana, S.Kom.,
M.Eng.Sc., OCA
Head of Department/Program

Has been checked and considered comply
to UMN standard
Date:



David Agustriawan, Ph.D
Ex-officio Informatics Study Program



COURSE PLANNING (RPKPS) UNIVERSITAS MULTIMEDIA NUSANTARA

COURSE NAME : Artificial Intelligence
 CODE / CREDIT : IF440 / 3
 SEMESTER : 3
 PREREQUISITE : EPM 101, CE 121, CE 319
 COURSE STATUS : Mandatory

A. COURSE DESCRIPTION

This course is a mandatory course given to undergraduate (S1) informatics students who are in semester 3. The focus of this course is theories related to artificial intelligence. The discussion emphasizes problem solving, knowledge representation, deduction reasoning, planning, aspects of uncertainty in thinking, learning and processing. soft computing. The learning process is carried out with active learning in the form of group discussions, and learning based on the results of studies conducted during one semester to achieve cognitive (level C3).

B. LEARNING OUTCOME

B.1. Program Expected Learning Outcomes (ELO) Related to the Course

IQF Level: 6

- ELO-2 Problem Solving and Critical Thinking: Students should be able to analyze complex problems, evaluate alternatives critically, and develop effective, innovative solutions using logical reasoning.;
- ELO-5 Intelligent Systems Proficiency: Students are expected to design, develop, and apply intelligent systems using AI and machine learning techniques to address practical challenges.

B.2. Course Learning Outcomes (CLO)

After passing this course, students will be able to use the skills and knowledge from this course as a AI Scientist in beginner level with competences as follows:

- ELO: 2 CLO-021 Students are able to master the applied scientific concepts of informatics – (C3)
- ELO: 5 CLO-051 Students can apply intelligent system using AI techniques to address challenges – (C3).

B.3. Course Sub Learning Outcomes (Sub-CLO)

- CLO-021 SubCLO-0211 Students are able to apply knowledge-based reasoning, probabilistic reasoning and soft computing (C3).
- CLO-051 SubCLO-0511 Students are able to conduct self-paced learning using the online learning platform in the field of artificial intelligence (C3).

C. LEARNING ANALYSIS

-Figure is attached-

D. TOPICS**1. Overview of Artificial Intelligence**

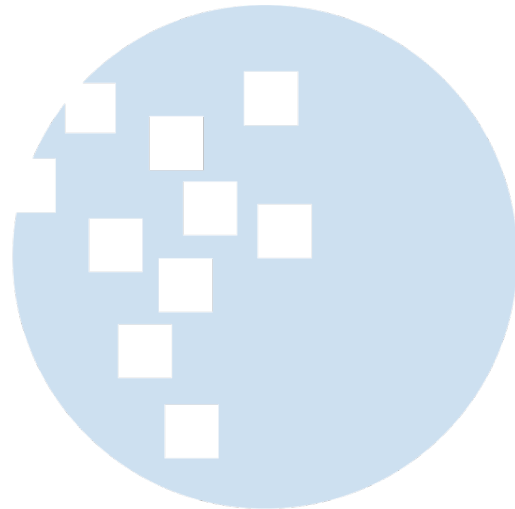
2. Models of knowledge representation and propositional logic
3. First-order Logic
4. Bayesian Theorems
5. Bayesian Networks
6. Markov Models
7. Hidden Markov Models

8. Overview of Machine Learning

9. Expectation-Maximization
10. Certainty Factors
11. Fuzzy Logic
12. Genetic Algorithm
13. Reinforcement Learning

14. Overview of Deep Learning**E. EVALUATION**

1. Attending lecture punctually is mandatory. Students will be considered absent if they are 15 minutes late.
2. Attending 14 lectures is mandatory. Attending minimum of 11 out of 14 meetings is required in order to take the final exam.
3. Final grade is determined by following components:
 - a. Midterm Exam : 25%
 - b. Final Exam : 25%
 - c. Activities :
 - Quizzes : 20%
 - Self-paced study (LMS) : 20%
 - Active participation : 10%



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Table: Assessment Distribution of Learning Outcomes

ELO	CLO	Sub-CLO	Assignments	Midterm Ex.	Final Ex.
ELO-02	CLO-021	SubCLO-0211 Students are able to apply knowledge-based reasoning, probabilistic reasoning and soft computing (C3).	30%	25%	25%
ELO-05	CLO-051	SubCLO-0511 Students are able to conduct self-paced learning using the online learning platform in the field of artificial intelligence (C3).	20%		

FINAL GRADING:

Score	Alphabetical Grade	Numerical Grade	Remarks
85 – 100	A	4	Excellent
80 – 84,99	A-	3,7	Good
75 – 79,99	B+	3,3	
70 – 74,99	B	3,0	
65 – 69,99	B-	2,7	Satisfactory
60 – 64,99	C+	2,3	
55 – 59,99	C	2,0	
45 – 54,99	D	1,0	Poor
0 – 44,99	E	0	Very Poor
	F	0	Academic Violation

F. REFERENCE AND RESOURCES**-Main-**

1. Russell, S., Peter Norvig (2021), Artificial Intelligence: A Modern Approach, 4th edition, New Jersey: Pearson Education.
2. Djong, A. M., & Widjaja, M. (2023). A Web-Based Expert System for Diagnosing Cervical Cancer Using Dempster-Shafer Method. 2023 7th International Conference on New Media Studies (CONMEDIA), 243–249. <https://doi.org/10.1109/CONMEDIA60526.2023.10428193>

-Additional/Supporting-

3. Do, C., Batzoglu, S. What is the expectation maximization algorithm?, Nat Biotechnol 26, 897–899 (2008). <https://doi.org/10.1038/nbt1406>
4. Ross, Timothy J. (2010), Fuzzy Logic with Engineering Applications, Edisi ke-3, United Kingdom: John Wiley & Sons.
5. Sutton, R. S., & Barto, A. G. (2018). Reinforcement Learning: An Introduction (2nd ed.). MIT Press. Retrieved from <http://incompleteideas.net/book/the-book-2nd.html>
6. Weber, R. (2011), Markov Chains, University of Cambridge. <http://www.statslab.cam.ac.uk/~rrw1/markov/M.pdf>

G. WEEKLY LESSON PLAN

Week	Course Sub-Learning Outcomes (Sub-CLO)	Topics & Sub-topics	Learning Methods and Activities	Timing	Assesment			Ref.
					Assessment type and Grading System	Indicators	Weight	
1.	Sub-CLO0511 Students are able to conduct self-paced learning using the online learning platform in the field of artificial intelligence (C3)	<u>Topics:</u> Overview of Artificial Intelligence <u>Sub-topics:</u> - Basic concepts of AI, - AI technologies and development history, - Application technologies and fields of AI, - AI development trends - AI applications in Indonesia	<u>Learning Methods:</u> Lectures <u>Activities:</u> Face to Face: Presentation of Material - Listen to the presentation of material about Basic concepts of AI, AI technologies and development history, Application technologies and fields of AI, AI development trends Discussion: - Discussion related to participants' perceptions regarding the Basic concepts of AI, AI technologies and development history, Application technologies and fields of AI, AI development trends E-Learning: - Accessing lecture materials. - Collect documents resulting from group discussions Task: - Independent Assignment: Students independently conduct a self-paced learning on the assigned learning platform.	2x50' 1x50' 2x60" 4x60'	<u>Assessment:</u> Progress in LMS <u>Grading system:</u> Progress in LMS	- The accuracy of students' perceptions is related to the concept of artificial intelligence - Completeness of various artificial intelligence approaches - Appropriateness of various artificial intelligence application technologies.	6%	
2.	Sub-CLO0211 Students are able to apply knowledge-based reasoning, probabilistic reasoning and soft computing.	<u>Topics:</u> - Knowledge representation model - Proposition Logic <u>Sub-topics:</u> - Knowledge representation model - Proposition Logic	<u>Learning Methods:</u> Lectures <u>Activities:</u> Face to Face: Presentation of Material - Listen to the presentation of material related to knowledge representation models and propositional logic - Discussion: - Work in groups to find a model of a proposition sentence E-Learning: - Accessing lecture materials. - Collect documents resulting from group work	2x50' 1x50'	<u>Assessment:</u> quiz 1 <u>Grading system:</u> Points per question	The accuracy of the model given for each logical sentence in the discussion group.		1:7,

Week	Course Sub-Learning Outcomes (Sub-CLO)	Topics & Sub-topics	Learning Methods and Activities	Timing	Assesment			Ref.
					Assessment type and Grading System	Indicators	Weight	
			Task: - Independent Assignment: - Students independently review the material presented in class. - Students independently conduct a self-paced learning on the assigned learning platform.	2x60' 4x60'				
3.	Sub-CLO0211 Students are able to apply knowledge-based reasoning, probabilistic reasoning and soft computing.	<u>Topics:</u> First Order Logic <u>Sub-topics:</u> - Difference between FOL and Propositional Logic - Syntax and Semantics of FOL - Resolution	<u>Learning Methods:</u> Lectures <u>Activities:</u> Face to Face: Presentation of Material - Listen to the presentation of material related to first-order logic-based knowledge representation models. Discussion: - Students use a resolution method based on the representation of knowledge in first order logical sentences. E-Learning: - Accessing lecture materials. - Collect documents resulting from group work Task: - Independent Assignment: - Students independently review the material presented in class. - Students independently conduct a self-paced learning on the assigned learning platform.	2x50' 1x50' 2x60' 4x60'	<u>Assessment:</u> Quiz 1 <u>Grading system:</u> Points per question	The accuracy of the calculation results from the resolution method based on the first order logic sentences that have been given in the discussion group		1:8-9
4.	Sub-CLO0211 Students are able to apply knowledge-based reasoning, probabilistic reasoning and soft computing.	<u>Topics:</u> - Bayesian Theorem <u>Sub-topics:</u> - Conditional & Joint Probability - Bayesian Theorem	<u>Learning Methods:</u> Lectures <u>Activities:</u> Face to Face: Presentation of Material - Listen to the presentation of material related to probability theory and Bayesian theorem Discussion: - Students carry out a classification process for a simple problem using a Bayesian Theorem based model. E-Learning:	2x50' 1x50'	<u>Assessment:</u> Quiz 1 <u>Grading system:</u> Points per question	The accuracy of the classification results is based on the calculation process of the Bayesian theorem based method based on group assignments	5%	1:12

Week	Course Sub-Learning Outcomes (Sub-CLO)	Topics & Sub-topics	Learning Methods and Activities	Timing	Assesment			Ref.
					Assessment type and Grading System	Indicators	Weight	
			- Accessing lecture materials. - Collect documents resulting from group work Task: - Independent Assignment: - Students independently review the material presented in class. - Students independently conduct a self-paced learning on the assigned learning platform.	2x60' 4x60'				
5.	Sub-CLO0211 Students are able to apply knowledge-based reasoning, probabilistic reasoning and soft computing.	<u>Topics:</u> - Bayesian Networks <u>Sub-topics:</u> - Bayesian Networks Diagram - Bayesian Networks Inference	<u>Learning Methods:</u> Lectures <u>Activities:</u> Face to Face: Presentation of Material - Listen to the presentation of material related to Bayesian networks theory Discussion: - Students carry out an inference process based on a given Bayesian networks diagram. E-Learning: - Accessing lecture materials. - Collect project proposals Task: - Independent Assignment: - Students independently review the material presented in class. - Students independently conduct a self-paced learning on the assigned learning platform.	2x50' 1x50' 2x60' 4x60'	<u>Assessment:</u> Quiz 2 <u>Grading system:</u> Points per question	Accuracy of calculation results based on the query and Bayesian Networks provided.		1:13
6.	Sub-CLO0211 Students are able to apply knowledge-based reasoning, probabilistic reasoning and soft computing.	<u>Topics:</u> - Markov Model <u>Sub-topics:</u> - Markov Chains - Transition Matrix - Applied Markov Chains	<u>Learning Methods:</u> Lectures <u>Activities:</u> Face to Face: Presentation of Material - Listen to the presentation of material related to Markov Chain and Transition Matrix Discussion: - Students carry out the calculation process based on the initial transition matrix or transition diagram provided. E-Learning:	2x50' 1x50'	<u>Assessment:</u> Quiz 2 <u>Grading system:</u> Points per question	Accuracy of calculation results based on questions given in group discussions.		6

Week	Course Sub-Learning Outcomes (Sub-CLO)	Topics & Sub-topics	Learning Methods and Activities	Timing	Assesment			Ref.
					Assessment type and Grading System	Indicators	Weight	
			- Accessing lecture materials. Task: - Independent Assignment: - Students independently review the material presented in class. - Students independently conduct a self-paced learning on the assigned learning platform.	2x60' 4x60'				
7.	Sub-CLO0211 Students are able to apply knowledge-based reasoning, probabilistic reasoning and soft computing.	<u>Topics:</u> - Hidden Markov Model (HMM) <u>Sub-topics:</u> - HMM Formalism - Viterbi Algorithm	<u>Learning Methods:</u> Lectures <u>Activities:</u> Face to Face: Presentation of Material - Listen to the presentation of material related to the Hidden Markov Chain model and the Viterbi algorithm Discussion: - Students formulate an HMM model for a problem and carry out the Viterbi algorithm calculation process based on the model they created. E-Learning: - Accessing lecture materials. - Collect documents resulting from group work Task: - Independent Assignment: - Students independently review the material presented in class. - Students independently conduct a self-paced learning on the assigned learning platform.	2x50' 1x50' 2x60' 4x60'	<u>Assessment:</u> Quiz 2 <u>Grading system:</u> Points per question	Accuracy of calculation results based on questions given in group discussions.	5%	1:14
Midterm Test Type or Assignment Methods: Essay Test (PL+FOL 5%, Bayes Theorem 5%, Bayesian Network 5%, Markov Chains 5%, HMM 5%)							25%	1
8.	Sub-CLO0511 Students are able to conduct self-paced learning using the online learning platform in the field of artificial intelligence (C3)	<u>Topics:</u> Overview of Machine Learning <u>Sub-topics:</u> - Machine Learning Algorithms, - Types of Machine Learning,	<u>Learning Methods:</u> Lectures <u>Activities:</u> Face to Face: Presentation of Material - Listen to the presentation of material about Machine Learning Algorithms, Types of Machine Learning, Machine Learning Process, Important Machine Learning Concepts, Common Machine Learning Algorithms.	2x50'	<u>Assessment:</u> Progress in LMS <u>Grading system:</u> Progress in LMS	- The accuracy of students' perceptions is related to the concept of machine learning - Completeness of various types of machine learning	6%	

Week	Course Sub-Learning Outcomes (Sub-CLO)	Topics & Sub-topics	Learning Methods and Activities	Timing	Assesment			Ref.
					Assessment type and Grading System	Indicators	Weight	
		- Machine Learning Process, - Important Machine Learning Concepts, - Common Machine Learning Algorithms	Discussion: - Discussion related to participants' perceptions regarding the Machine Learning Algorithms, Types of Machine Learning, Machine Learning Process, Important Machine Learning Concepts, Common Machine Learning Algorithms E-Learning: - Accessing lecture materials. - Collect documents resulting from group discussions Task: - Independent Assignment: Students independently conduct a self-paced learning on the assigned learning platform.	1x50' 2x60" 4x60'		- Appropriateness of common machine learning algorithms		
9.	Sub-CLO0211 Students are able to apply knowledge-based reasoning, probabilistic reasoning and soft computing.	<u>Topics:</u> - Expectation Maximization Algorithm <u>Sub-topics:</u> - Maximizing Likelihood Estimations (MLE) - Optimization Problems - EM Algorithm	<u>Learning Methods:</u> Lectures <u>Activities:</u> Face to Face: Presentation of Material - Listen to the presentation of material related to MLE problems and EM algorithms Discussion: - Students carry out a calculation process based on the questions given. E-Learning: - Accessing lecture materials. Task: - Independent Assignment: - Students independently review the material presented in class. - Students independently conduct a self-paced learning on the assigned learning platform.	2x50' 1x50' 2x60' 4x60'	Assessment: Quiz 3 Grading system: Points per question	Accuracy of calculation results based on questions given in group discussions.		1:21, 3
10.	Sub-CLO0211 Students are able to apply knowledge-based reasoning, probabilistic reasoning and soft computing.	<u>Topics:</u> - Certainty Factor <u>Sub-topics:</u> - Evidential reasoning - CF Formalism - Dempster-Shafer Theory	<u>Learning Methods:</u> Lectures <u>Activities:</u> Face to Face: Presentation of Material - Listen to the presentation of material related to Certainty Factor Discussion: - Students carry out a calculation process based on an example problem given E-Learning: - Accessing lecture materials.	2x50' 1x50'	Assessment: Quiz 3 Grading system: Points per question	Accuracy of calculation results based on questions given in group discussions.	5%	2

Week	Course Sub-Learning Outcomes (Sub-CLO)	Topics & Sub-topics	Learning Methods and Activities	Timing	Assesment			Ref.
					Assessment type and Grading System	Indicators	Weight	
		- Example of DS method in healthcare (SDG 3)	Task: - Independent Assignment: - Students independently review the material presented in class. - Students independently conduct a self-paced learning on the assigned learning platform.	2x60' 4x60'				
11.	Sub-CLO0211 Students are able to apply knowledge-based reasoning, probabilistic reasoning and soft computing.	<u>Topics:</u> - Fuzzy Logic <u>Sub-topics:</u> - Crisp Sets - Fuzzy Set Theory - Membership Functions - Fuzzy Rules - Proses Fuzzification & Defuzzification - Fuzzy Inference system - Exercise	<u>Learning Methods:</u> Lectures <u>Activities:</u> Face to Face: Presentation of Material - Listen to the presentation of material related to Fuzzy logic Discussion: - Students carry out a calculation process based on an example problem given E-Learning: - Accessing lecture materials. Task: - Independent Assignment: - Students independently review the material presented in class. - Students independently conduct a self-paced learning on the assigned learning platform.	2x50' 1x50' 2x60' 4x60'	<u>Assessment:</u> Quiz 3 <u>Grading system:</u> Points per question	Accuracy of calculation results based on questions given in group discussions.		4:2, 4:4, 4:5
12.	Sub-CLO0211 Students are able to apply knowledge-based reasoning, probabilistic reasoning and soft computing.	<u>Topics:</u> - Genetic Algorithm <u>Sub-topics:</u> - Basic Concepts of GA - Representation Techniques - Fitness Function Design - Selection Methods - Crossover Techniques - Mutation Operators - Applications of GA	<u>Learning Methods:</u> Lectures <u>Activities:</u> Face to Face: Presentation of Material - Listen to presentations on material related to Genetic algorithm Discussion: - Students carry out manual calculation processes from a multi-layer network model to study a learning task E-Learning: - Accessing lecture materials. - Collect documents resulting from group work Task: - Independent Assignment:	2x50' 1x50' 2x60'	<u>Assessment:</u> Quiz 4 <u>Grading system:</u> Points per question	Accuracy of calculation results based on questions given in group discussions.		1:4

Week	Course Sub-Learning Outcomes (Sub-CLO)	Topics & Sub-topics	Learning Methods and Activities	Timing	Assesment			Ref.
					Assessment type and Grading System	Indicators	Weight	
			- Students independently review the material presented in class. - Students independently conduct a self-paced learning on the assigned learning platform.	4x60'				
13.	Sub-CLO0211 Students are able to apply knowledge-based reasoning, probabilistic reasoning and soft computing.	<u>Topics:</u> - Reinforcement Learning <u>Sub-topics:</u> - Basic concepts of RL - Types of RL - Value-Based Methods - Policy-Based Methods - Exploration Strategies	<u>Learning Methods:</u> Lectures <u>Activities:</u> Face to Face: Presentation of Material - Listen to presentations on material related to Reinforcement Learning Discussion: - Students carry out manual calculation processes of RL. E-Learning: - Accessing lecture materials. - Collect documents resulting from group work Task: - Independent Assignment: - Students independently review the material presented in class. - Students independently conduct a self-paced learning on the assigned learning platform.	2x50' 1x50' 2x60' 4x60'	<u>Assessment:</u> Quiz 4 <u>Grading system:</u> Points per question	Accuracy of calculation results based on questions given in group discussions.	5%	1:23, 5:6
14.	Sub-CLO0511 Students are able to conduct self-paced learning using the online learning platform in the field of artificial intelligence (C3)	<u>Topics:</u> Overview of Deep Learning <u>Sub-topics:</u> - Deep Learning, - Training Rules, - Activation Functions, - Normalization, - Optimizers, - Neural Network Types	<u>Learning Methods:</u> Lectures <u>Activities:</u> Face to Face: Presentation of Material - Listen to the presentation of material about Deep Learning, Training Rules, Activation Functions, Normalization, Optimizers, Neural Network Types Discussion: - Discussion related to participants' perceptions regarding the Deep Learning, Training Rules, Activation Functions, Normalization, Optimizers, Neural Network Types E-Learning: - Accessing lecture materials. - Collect documents resulting from group discussions Task:	2x50' 1x50' 2x60"	<u>Assessment:</u> Progress in LMS <u>Grading system:</u> Progress in LMS	- The accuracy of students' perceptions is related to the concept of deep learning - Completeness of various components of deep learning - Appropriateness of deep learning algorithms	8%	

Week	Course Sub-Learning Outcomes (Sub-CLO)	Topics & Sub-topics	Learning Methods and Activities	Timing	Assesment			Ref.
					Assessment type and Grading System	Indicators	Weight	
			- Independent Assignment: Students independently conduct a self-paced learning on the assigned learning platform.	x60'				
Final Test Type or Assignment Methods: Essay Test (EM 5%, CF 5%, FL 5%, GA 5%, RL 5%)							25%	1, 2, 3,4,5

Task/Project Details:**1. Task**

Course	:	Artificial Intelligence	Course Code	:	IF440
Task/Project Name	:	Self-paced online learning HCIA-AI V3.5	Weight	:	20%
Related Sub-CLO	:	0511			
A. Individual Activities					
Description	:	Students are expected to demonstrate the ability to conduct a self-paced learning.			
Assessment Type	:	Formative assessment			
Output and format	:	Output: Completed the online course and passed the exam Format: n/a			
Indicators, Criteria, and Weight	:	a. Learning progress (0-100%): 15% (completed the online course with minimum learning duration of 10 hours) b. Passed the online exam: 5% (600/1000)			

2. Task

Course	:	Artificial Intelligence	Course Code	:	IF440
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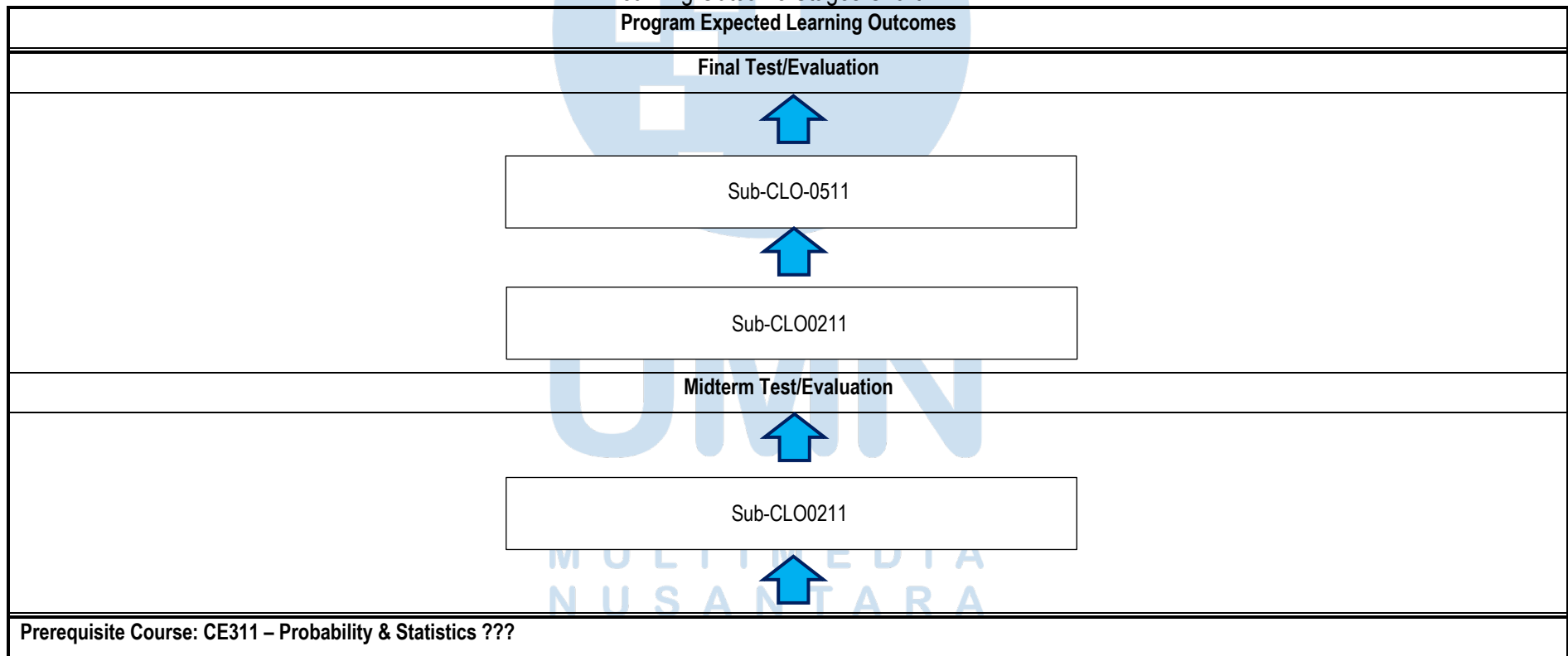
Task/Project Name	: Quizzes	Weight	: 20%
Related Sub-CLO	: 0211		
B. Individual Activities			
Description	Students are able to apply knowledge-based reasoning, probabilistic reasoning and soft computing.		
Assessment Type	Formative assessment		
Output and format	Output: Quiz in elearning Format: n/a		
Indicators, Criteria, and Weight	a. Correct answers		

3. Task

Course	: Artificial Intelligence	Course Code	: IF440
Task/Project Name	: Active Participation	Weight	: 10%
Related Sub-CLO	: 0211		
C. Individual Activities			
Description	Students are able to apply knowledge-based reasoning, probabilistic reasoning and soft computing.		
Assessment Type	Formative assessment		
Output and format	Output: Assignment in elearning Format: n/a		

Indicators, Criteria, and Weight	a. Pre-class assignment (50%) b. Active participation in class (asking/answering questions) (50%) a. Absence: 1 point b. Presence: 2 points c. Presence and actively asking/answering questions: 4 points
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Attachment: Learning Outcome Analysis



H. Revision History

Course Code	Revision No	Date in Effect	Changes
IF534	1	08-08-2023	Changes to the new RPKPS format
IF440	2	23-01-2024	Adding the online course
IF440	3	22-08-2025	Update ELO and topics

